

Midterm 2018, “The Mathematical Foundations of Political Methodology”

You have 120 minutes for this exam. You may use a calculator and refer to your Pemberton & Rau book. Please do not use the internet during the exam. Show your work.

1. Consider the following sets of real numbers:

$$A = \{x : 1 \leq x \leq 5\}$$

$$B = \{x : 0 \leq x \leq 2\}$$

$$C = \{x : x < 0\}$$

Describe each of the following sets of real numbers in words

(a) A^c

(b) $A \cup B$

(c) $B \cap C^c$

(d) $(A \cup B) \cap C$

(e) $(A \cup B)^c \cap C$

2. Find the limit of each of the following sequences as $n \rightarrow \infty$, if the limit exists.

(a)

$$\frac{4n}{(n^2 - 7)}$$

(b)

$$\frac{4n^{\frac{1}{2}}}{(n^2 - 7)} + 1$$

(c)

$$\frac{4n}{(n^{\frac{1}{2}} - 7)} + 1$$

3. Use the (δ, ϵ) definition of a limit to prove that

$$\lim_{x \rightarrow 3} 4x - 11 = 1$$

4. Use the definitions of continuity and differentiability to evaluate whether or not the following function is 1) continuous and 2) differentiable at $x = 0$.

$$f(x) = x^{\frac{1}{3}}$$

5. Let

$$f(x) = \frac{x}{1 + e^x}$$

- (a) Is $f(x)$ concave, convex or both?
(b) Find any asymptotes that f may have.
6. Perform the following matrix operations:

- (a) Find the inverse of

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

- (b) Find the rank of

$$A = \begin{bmatrix} 3 & 1 & 1 & 2 \\ 2 & 1 & -1 & 2 \\ 0 & 3 & 2 & 0 \\ 1 & 0 & 3 & 0 \end{bmatrix}$$

- (c) What x would make the following matrix singular?

$$\begin{bmatrix} 1 & 3 \\ x & 2 \end{bmatrix}$$

7. If A and B are matrices, which of the following imply the others:

- (a) $AB=BA$
(b) Either A or B is the identity matrix.
(c) A is the inverse of B.
(d) A and B are symmetric.

8. What constant k solves the following equation?

$$\int_0^{10} ky^2 dy = 1$$

9. What values of the constant a satisfies the following equation?

$$\int_0^a \frac{2x}{1+x^2} dx = 1$$